INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
END-SPRING SEMESTER EXAMINATION 2024-25

Subject No. : CS60075 **Subject : Natural Language Processing**

Duration: 3 Hours **Full Marks: 80**

Department: Computer Science and Engineering

Specific charts, graph paper, log book etc., required: NA

Special Instructions :

- *Attempt all questions.* ***All parts of the same question must be answered together.***
- *Answers should be brief and to-the-point. Sketchy answers will be penalized.*
- *Make reasonable assumptions only if necessary, and state any assumptions made. No clarification can be provided in the examination hall.*
- *All working steps must be shown. Writing the answer without showing steps will not be given any credit, even if the answer is correct. You can use calculators.*

Question 1 (Miscellaneous) [10 x 2 = 20]

State whether the following statements are True or False, giving a short (1-2 sentences) meaningful explanation or counter-example. Marks will be awarded only if both the True/False label and the explanation are correct.

(a) Different hidden state parameters are used in an RNN for each time step in the input sequence.
(b) The "forget gate" of a GRU is used to decide how much of the previous hidden state should be retained.
(c) Bidirectional RNNs use completely separate parameters for each direction (forward and backward) and share no information between them.
(d) Neural Machine Translation (NMT) models can handle long sentences better than Statistical Machine Translation models.
(e) In the Seq2Seq architecture, the decoder gets access to only the final context vector generated by the encoder from the input.
(f) The attention mechanism ensures that the decoder treats all input words equally during decoding.
(g) Positional encodings are used in the Transformer to indicate the relative sequence of words in the input.
(h) The same self-attention mechanism is used in the encoder and decoder layers of a Transformer.
(i) Pre-trained models usually require much less fine-tuning data, compared to traditional supervised models.
(j) BERT is trained to predict the next word in a sentence.

Question 2 (Word embeddings, RNNs) [4 + 6 + 6 + (2 + 2) = 20]

(a) Suppose we have the following word vectors: *play* = [1, 0, 0]; *ring* = [1, 1, 0]; *catch* = [0, 0, 2]. Now we get two new examples of a new word *qep: "qep play" and "qep catch"*. We are going to learn parameters for *qep* without changing the rest of our vector space. Which context embedding out of the following gives the best value for *qep* based on the two given examples? Give a proper explanation.

I. [10, 0, 5] *II.* [10, -10, 0] *III.* [10, -10, 5] *IV.* [0, -10, 5]

(b) The following diagram depicts a single RNN unit, where x_t, h_{t-1}, h_t and y_t are all scalars. The hidden unit h_t has an initial value of 0 (at time t = 0).

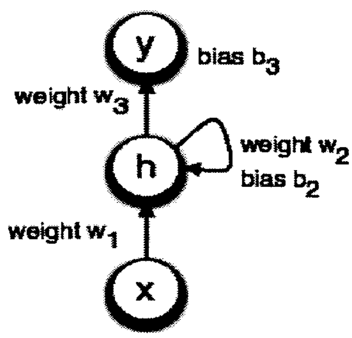
w_1 = ________

w_2 = ________

b_2 = ________

w_3 = ________

b_3 = ________



Suppose that *f* and *g* are activation functions defined as follows:
$f(x) = 0$ if $x < 0$ and $f(x) = 1$ if $x \geq 0$
$g(x) = x$
and assume that for RNNs: $h_t = f(w_1 x_t + w_2 h_{t-1} + b_2)$ and $y_t = g(w_3 h_t + b_3)$

Fill in weights (w_1, w_2, w_3), and biases (b_2, b_3) so that the RNN initially outputs 0, but as soon as it receives an input of 1, it switches to output 1 for all subsequent time steps. Assume that the input sequence always starts with 0. For instance, the input 000101001 should produce the output sequence 000111111. Provide the equations you use to get the answers. No marks will be provided unless proper explanations and equations are given.

(c) Calculate the number of *learnable parameters* of a LSTM cell where the hidden state is of size *h =100* and the input vector size is *x = 32.* Show all calculations. The equations for a LSTM cell are given below, where the symbols have the usual meaning as discussed in class.

$$i_t = \sigma(W_i h_{t-1} + U_i x_t + b_i)$$
$$f_t = \sigma(W_f h_{t-1} + U_f x_t + b_f)$$
$$o_t = \sigma(W_o h_{t-1} + U_o x_t + b_o)$$
$$\widetilde{c}_t = tanh(W h_{t-1} + U x_t + b)$$
$$c_t = f_t \circ c_{t-1} + i_t \circ \widetilde{c}_t$$
$$h_t = o_t \circ tanh(c_t)$$
$$y_t = h_t$$

(d) An LSTM has the following values (all of size 2) at timestep t:
Forget gate = [1, 0.5]
Input gate = [0, 0.3]
Candidate cell state = [0.7, 0.9]
Previous cell state = [1, 2]
output gate = [0.8, 0.6]
Compute the new cell state c_t and the new hidden state h_t. Consider that *tanh* is approximated by element-wise function: tanh(z) ≈ z / (1 + |z|). Show the calculations.

Question 3 (Seq2seq, cross attention) [(4+2) + 8 + 3 + 3 = 20]

(a) You are using an RNN-based sequence-to-sequence framework with simple dot-product attention. Suppose that the encoder-side hidden state representations are [1, 1], [2, 1], [-1, 0], [1, -1] at different time steps. If the decoder hidden state at a given time is [-1, 2], what will be the attention-weighted encoder representation at this time step? In terms of alignment with the decoder hidden state, how can we say that this attention-weighted representation is better than the mean of all encoder hidden state representations? Justify numerically.

(b) You have a bi-gram language model that has been trained on the following probability transition matrix (cell [i, j] is probability of transition from token i to j).

	<START>	nlp	is	fun	<END>
<START>	0	0.4	0.2	0.3	0.1
nlp	0	0.1	0.5	0.2	0.2
is	0	0.4	0.1	0.4	0.1
fun	0	0.2	0.4	0.1	0.3
<END>	0	0	0	0	0

During decoding, you start with the <START> token at time t = 0. Perform a beam search (with beam width = 2) up to four time steps (t = 1, 2, 3, 4). Show all steps of the calculation. You must also show all branches of the calculation. If some branch terminates (<END> generated), continue with the other branches. What are the terminated strings obtained within t = 4? What are the scores (sum of log probabilities) for these strings?

(c) Why is an RNN based seq2seq model not expected to work well when the source sentence is too long? Explain how the attention model helps to get better translation for longer sentences.

(d) Suppose you are using BLEU score to evaluate three candidate dialog generations A: "rained last night", B: "it rained", C: "it rained yesterday" for the ground truth "it rained last night". What will be the ordering of BLEU scores for A, B and C if you use only unigrams for the calculations? Justify your answer briefly.

Question 4 (Transformers, self attention, pre-training) [8 + 6 + (2 + 2 + 2) = 20]

(a) Consider a simplified one-layer Transformer architecture with a single attention head. Suppose that this model only accepts input sequences up to 8 tokens long. The input sequence to be processed is "computer science engineering". The initial word embeddings are:

$\boldsymbol{x}_{\text{computer}} = [1\ 0\ 1]$ $\boldsymbol{x}_{\text{science}} = [0\ 0\ 1]$ $\boldsymbol{x}_{\text{engineering}} = [1\ 1\ 0]$

For encoding positionality, 3-bit positional embeddings are added to the initial word embeddings. These positional embeddings are [0 0 0], [0 0 1], [0 1 0], [0 1 1], [1 0 0], [1 0 1], [1 1 0] and [1 1 1] for each of the 8 permissible sequence positions.

The Q, K and V projection matrices are as follows:

$$\boldsymbol{W}^{Q} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix} \qquad \boldsymbol{W}^{K} = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{bmatrix} \qquad \boldsymbol{W}^{V} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

For scoring, use standard dot product (no scaling). For attention-weighting, instead of softmax, use the following function $\lambda_j = \lambda_j / \sum \lambda_j$. Calculate the output vectors of the self-attention layer for each of the three words. Round the values in the final answer to two decimal places.

(b) Consider a layer-norm (with residual connections) and a single feed-forward network (without any bias or activation) on top of the self-attention layer described in (a). The weights of the feed-forward network are given below. Assume there is no layer-norm or residual connection on top of the feed-forward layer. What will be the output vectors for each of the three words after passing through this setup? Round the values in the final answer to two decimal places. Also, when starting your calculations, use the rounded values obtained in (a).

$$\boldsymbol{W}^{FFNN} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

(c) Answer the following questions briefly:

(i) Why are residual connections important in Transformers?

(ii) Describe how tokens are masked for BERT pre-training.

(iii) What are the differences between Encoder and Decoder blocks in Transformer?